

# Turbofan and Turboprop Engine Trade Study

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## Introduction:

Determining the most effective engine type is crucial in aviation and aircraft design as it influences the performance of flight, efficiency of use, costs, and other critical aspects. Although these widely implemented engine types, turboprop, and turbofan may share design details, they are prominently different in terms of functionality and purpose. Thus, implementing a trade study is an effective way of determining which engine is best suited for our prompt of creating an 80-passenger aircraft. Comparing and evaluating the characteristics of the two engine types will ultimately enable a more accurate and educated conclusion as to which engine will be of better effective and efficient for the prompt.

## Cost Efficiency:

When determining the cost efficiency of both turboprop and turbofan engines, it is crucial to consider multiple causes and contributing factors to the costs of maintaining each type of engine. These factors include the complexity of the mechanics of the engine, as well as fuel consumption. A study conducted by the Aircraft Owners and Pilots Association (AOPA) repeatedly showed that engines with similar traits to turbofans and turbofans themselves are more costly than turboprop engines. The article explains this by relating to the fact that turbofans have much more complex engineering and thus, make maintenance longer and more difficult, in turn leading to higher costs overall (Marsh, 2016). Ultimately, the cost efficiency of an engine is a prominent factor in determining which engine is better suited for a given circumstance. Highlighting the cost differences between the two engines enables clearer comparisons between the engines, which is significantly beneficial to conducting a trade study.

### Fuel Efficiency:

One of the key advantages of turboprop engines is their efficiency in fuel consumption. According to the 14-Day Pilot, turboprop engines excel at low-altitude, short-distance flights, which in turn contributes to low fuel usage (Febrian, 2024). The effective performance in intended conditions and aspects allows for minimal fuel usage, ultimately extending the aircraft's flight time and improving overall flight performance. Conversely, turbofans generally have lower fuel efficiency, as aircraft with these engines are usually intended for long-distance, high-altitude flights, which in turn leads to decreased fuel efficiency with use (Febrian, 2024). Although turbofans are most efficient in their sector of high altitude and extensive flights, they have significantly higher fuel usage than turboprop engines.

### Safety:

Turboprop and Turbofan are both extremely safe and reliable when working with installations or maintenance exercises. However, the unsafe characteristics of the engines mainly stem from their installation setup prerequisites and the wear that occurs on the engines over substantial use. First, by analyzing the turbofan engine, Airplane Academy states that one of the safety benefits of the turbofan engine is that its bypass air significantly decreases the risk of icing and as well as pressurization (Gasmire, n.d.). Icing and pressurization inside the engine can lead to engine stall, and in extreme cases, engine failure. The article also states that turbofan engines are less prone to asymmetric thrust. This can decrease safety by increasing the difficulty of level flight. Conversely, turboprop engines' propellers hinder the range of adjustment of the engine. Furthermore, Airplane Academy states that turboprop engines' propellers can make it difficult to recover from engine failure due to their limited range in positioning on the aircraft wing. Moreover, it is also stated that turboprop propellers generate a significant amount of torque, which can result in rollovers of the aircraft (Gasmire, n.d.). This

difficulty in effectively recovering the engine from failure and the more significant change of rollover pose a danger to the passengers and pilots within the aircraft.

#### Comfort:

The comfort of an aviation aircraft can be left to the discussion, as perspectives could argue should not be considered and can be sacrificed for other aspects, such as engine efficiency and maintenance, while opposing perspectives could argue that comfort should always be considered when implementing an engine. Although the prioritization of comfort can be argued and varies depending on the purpose of the aircraft, developing a trade study that acknowledges all aspects of engines and their impact on aviation is crucial to forming the clearest decision. The noise created by an engine is a primary contributor to the comfort of flying. The structure of turbofan engines results in less noise being created compared to turboprop engines due to the exposed propeller. A study comparing the noise output of two similar-sized aircraft determined that the turbofan engine aircraft, the Bombardier CRJ 900, produces less cabin noise than the turboprop aircraft, the ATR 72-500 (Lee, 2023). Ultimately, the less noise produced by the turbofan leads to the conclusion that turbofan engines lead to more concentrated flying conditions for pilots and more comfortable conditions for passengers.

#### Maintenance:

Determining the benefits and deficits behind the maintenance of the two aircraft engines is crucial in their comparison and is primarily comprised of the mechanical complexity of the engine. Pilot Mall explains that turboprop engines tend to be more reliable as they have fewer moving parts and result in less extensive maintenance. On the other hand, the mechanical complexity of turbofan engines leads to the need for higher levels of maintenance to keep the engine in operating condition (Glazer, 2023). This difference in design ultimately leads to

turboprop engines being more reliable and less prone to mechanical difficulties, resulting in lower maintenance costs and time.

Evaluation Criteria Scoring:

Cost: Red: More expensive - Yellow: Equally expensive - Green: Less expensive

Fuel Efficiency: Red: More efficient - Yellow: Equally efficient - Green: Less efficient

Safety: Red: Lower safety - Yellow: Equivalent Safety - Green: Higher safety

Comfort: Red: Less comfortable - Yellow: Equally comfortable - Green: More comfortable

Maintenance: Red: High maintenance - Yellow: Equal maintenance - Green: Low maintenance

| <i>Evaluation<br/>Criteria Table</i> | <b>Criteria</b> |                    |        |         |             |
|--------------------------------------|-----------------|--------------------|--------|---------|-------------|
| <b>Options</b>                       | Cost            | Fuel<br>Efficiency | Safety | Comfort | Maintenance |
| Turbofan                             |                 |                    |        |         |             |
| Turboprop                            |                 |                    |        |         |             |

Conclusion:

Ultimately, both the turbofan and turboprop engines proved to have unique advantages and disadvantages and showed their superiority in some fields over others. Moreover, the intended use of the aircraft on which the engines are implemented also has a prominent role in determining each engine's performance and usefulness. The turboprop showed its effectiveness and high performance in shorter distances and lower-altitude flights. Conversely, the turbofan was shown to be both more effective and efficient in costs and fuel consumption with longer distance and higher altitude flights. Examining the strengths of the turbofan and turboprop

engines, the turboprop not only exceeds the turbofan in more categories of the criteria but also indicates a higher level of consistency and reliability. Overall, the 80-passenger plane should also implement a turboprop engine as opposed to a turbofan.

## References

- Febrian, I. (2024, October 30). *Aircraft engines battle: Piston vs turboProp vs turboFan*. 14 Day Pilot. Retrieved January 4, 2025, from <https://14daypilot.com/aviator-blog/read/aircraft-engines-battle-piston-vs-turboprop-vs-turbofan>
- Gasmire, C. (n.d.). *Turboprop vs. turbofan: Safety, efficiency, and performance*. Airplane Academy. Retrieved January 4, 2025, from <https://airplaneacademy.com/turboprop-vs-turbofan-safety-efficiency-and-performance/>
- Glazer, N. (2023, July 5). *Turboprop vs jet: Learn their advantages and disadvantages*. Pilot Mall. Retrieved January 4, 2025, from <https://www.pilotmall.com/blogs/news/turboprop-vs-jet-learn-their-advantages-and-disadvantages#:~:text=Fewer%20moving%20parts%20in%20a,hourly%20operation%20cost%20is%20lower.>
- Lee, H. M., Garg, S., Lim, K. M., & Lee, H. P. (2023). Comparison of cabin noise of turboprop and turbofan aircrafts. *Applied Acoustics*, 214, 109651. <https://doi.org/10.1016/j.apacoust.2023.109651>
- Marsh, A. K. (2016, October 31). *Operating costs for jets, turboprops detailed*. Aircraft Owners and Pilots Association. Retrieved January 4, 2025, from <https://www.aopa.org/news-and-media/all-news/2016/october/31/hourly-operating-costs>